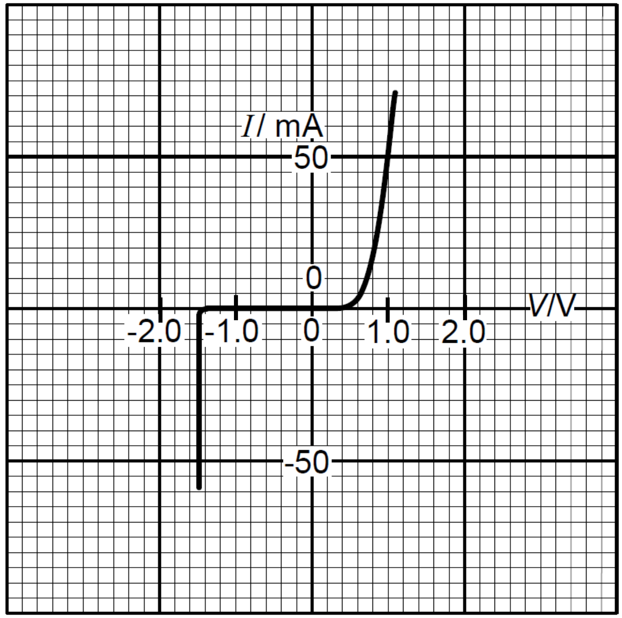
22 The variation with potential difference V of the current I in a semiconductor diode is shown below.



What is the resistance of the diode for applied potential differences of +1.0 V and –1.0 V?

	+1.0 V	–1.0 V
A	20 Ω	infinite
B	20 Ω	zero
C	0.05 Ω	infinite
D	0.05 Ω	zero

Ans: A – at -1.0 V, the resistance must be very large so that the current is zero. At +1.0 V, $R = V/I = 1/(50 \times 10^{-3}) = 20\ \Omega$